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# 1 Introduction

The **Statistics Center** is developed by **Bosch Power Tools**. If you have any question or feedback, you can write an E-Mail to [daniel.vesenmaier@de.bosch.com](mailto:daniel.vesenmaier@de.bosch.com) or [matheus.moraes@br.bosch.com](mailto:matheus.moraes@br.bosch.com)

Disclaimer: Although the Statistics Center has been carefully tested, we cannot guarantee that it is completely free of errors. Use of this tool is at your own risk.

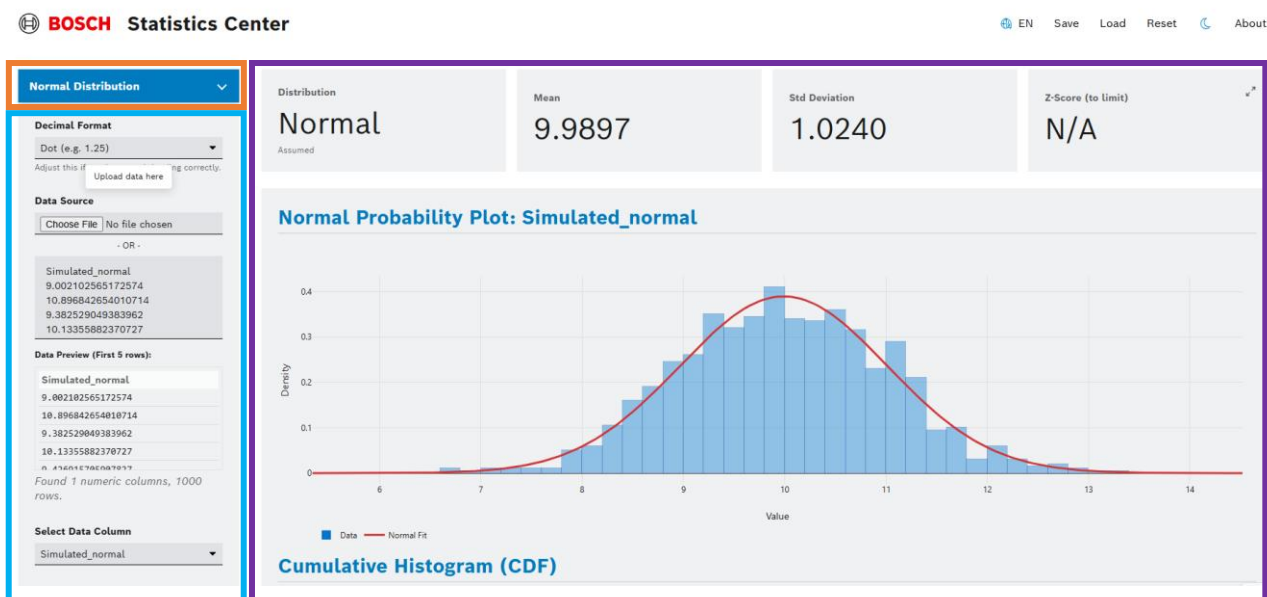
The Statistics Center is a browser-based application for statistical analysis that runs as a static webpage. The tool provides a suite of interactive modules for statistical analysis and visualization designed for development, test and quality engineers.

# 2 User Interface

The application is divided into two main areas:

- **Left Sidebar (Control Panel):** This is where you input your data and configure the parameters for the selected analysis.
- **Right Panel (Results Panel):** This is where the results of your analysis are displayed in the form of charts and tables.
- **Module Selection**

Use the dropdown menu at the top left to select the desired analysis module (e.g., "Process Capability," "Regression Analysis," etc.). The options available in the sidebar will change depending on the selected module.



## 3 Data Input

You can load data into the application in three ways:

1. **Paste Data:** Copy your data (e.g., from Excel) and paste it directly into the "Data Source" text area.
2. **File Upload:** Click the "Choose File" button to upload a **.csv**, **.txt**, **.xlsx**, or **.xls** file from your computer.
3. **Monte Carlo Generation:** Use the "Monte Carlo" module to generate simulated data based on statistical distributions.

**Decimal Format:** Ensure you select the correct **Decimal Format** (Period or Comma) so that your numerical data is read correctly.

## 4 Analysis Modules

Here is a detailed description of each analysis module.

### 4.1 Normal Distribution

**Purpose:** To specifically test if your data follows a normal distribution. This is often a critical assumption for many other statistical analyses, such as Process Capability and Hypothesis Tests.

**How to Use:**

- Choose the data column you wish to test.
- Optionally, enter an upper or lower data bound to estimate the percentage of data that would fall outside this limit.
- Click on "Analyze".

**Output:** The analysis provides a Normal Probability Plot and a goodness-of-fit test (e.g., Anderson-Darling) with a corresponding p-value. A high p-value (typically  $> 0.05$ ) suggests that the data is normally distributed.

### 4.2 Multi-Distribution

**Purpose:** To fit several different statistical distributions (e.g., Normal, Lognormal, Weibull, Exponential) to your data simultaneously to determine which one provides the best fit.

**How to Use:**

- Select the data column to be analyzed.
- Optionally, enter an upper or lower data bound to estimate the percentage of data that would fall outside this limit.
- Click on "Analyze".

**Output:** A ranked list of the best-fitting distributions, ordered by a goodness-of-fit statistic. The output includes the estimated parameters for each distribution and a chart showing the distribution curves overlaid on a histogram of your data for visual comparison.

## 4.3 Descriptive Statistics

**Purpose:** Calculate basic statistical metrics and create common statistical graphs.

**How to Use:**

- Select one or more "Numeric Columns" to analyze.
- Optionally, select a "Group By" column to separate results by category.
- Select the statistics and chart types you want to display.
- Click on "Analyze".

**Output:** A table of statistics (mean, standard deviation, etc.) and the selected charts (Histogram, Box Plot, Time Series Plot).

## 4.4 Monte Carlo Simulation

**Purpose:** Generate random data from various statistical distributions.

**How to Use:**

1. Select a **Distribution** (Normal, Lognormal, Weibull).
2. Enter the parameters for the selected distribution.
3. Specify the **Sample Size (N)**.
4. Click "Generate Data".

**Output:** A histogram and a CDF plot of the generated data. You can then load this data into the application as a new data source.

## 4.5 Pareto Analysis

**Purpose:** Identify the most significant factors in a dataset (the "vital few").

**How to Use:**

- Select a **Category Column**.
- Optionally, select a **Value Column** to sum values instead of just counting frequency.
- Click on "Update graph".

**Output:** A Pareto chart that sorts the categories from largest to smallest, and a table with the corresponding frequencies, percentages, and cumulative percentages.

## 4.6 Capability

**Purpose:** Assess whether your process is capable of producing results within specification limits.

**How to Use:**

- Select a data column.
- Enter the **Lower Specification Limit (LSL)** and/or the **Upper Specification Limit (USL)**.
- Specify the **Subgroup size** to estimate the "within" variation (1 for individual values).
- Click on "Analyze".

**Output:** A comprehensive report with capability indices (Cp, Cpk, Pp, Ppk, Cpm) and expected PPM (Parts Per Million) failure rates. A chart visualizes the data relative to the specification limits.

## 4.7 Control Charts

**Purpose:** Monitor the stability of a process over time.

**How to Use:**

- Select the **Control Chart Type:** "Individuals & Moving Range Chart (I-MR)" or "X-bar & Range Chart (Xbar-R)".
- For Xbar-R charts, select a **Subgroup Column**.
- Select the **Alarm Rules** to apply.

**Output:** Control charts showing the data points relative to the calculated control limits. A table lists all points that violate the selected alarm rules.

## 4.8 MSA / Gage R&R

**Purpose:** Evaluate the precision and variability of your measurement system.

**How to Use:**

- Select the columns for **Measurement**, **Part ID**, and **Operator ID**.
- Choose the **Method** (ANOVA or Mean and Range).
- Optionally, enter the **Process Tolerance** for additional calculations.
- Click on "Analyze".

**Output:** A table of variance components (%Contribution, %Study Variation), the Number of Distinct Categories (ndc), and charts to visualize the sources of variation.

## 4.9 Tolerance Interval

**Purpose:** Calculate limits that cover a specified proportion of the population with a certain confidence level.

**How to Use:**

- Select a data column.
- Set the **Confidence Level** and **Population Coverage**.
- Click on "Analyze".

**Output:** A table showing the tolerance limits for both the normal distribution method and the non-parametric method, along with a normality test to determine which method is more appropriate.

## 4.10 Hypothesis Tests

**Purpose:** Make statistical decisions about population parameters based on sample data.

**How to Use:**

- Select the **Test Type** (e.g., 1-Sample t-Test, ANOVA).
- Select the appropriate **Response** and **Factor columns**.
- Enter a **Target Mean** if required.
- Click on “Analyze”

**Output:** A summary of the test with the p-value, a table of group statistics, and a box plot for visually comparing the groups.

## 4.11 Regression & Correlation

**Purpose:** Model the relationship between a dependent variable (Y) and one or more independent variables (X).

**How to Use:**

- Select the **Response Variable (Y)**.
- Select one or more **Explanatory Variables (X)**.
- Select the **Model Type** (Linear, Quadratic).
- Click on “Analyze”.

**Output:** An ANOVA table, coefficients, R-squared values, and the regression equation. Diagnostic plots like a normal probability plot of residuals and a contour plot are also created.

## 4.12 Design of Experiments (DOE)

**Purpose:** Create and analyze 2-level full factorial experiments.

**How to Use:**

- Define your **Factors** and their low/high levels.
- Click "Generate Design" to create the experimental matrix.
- Enter the **Response (Y) values** for each run in the table.
- Click "Analyze Experiment".

**Output:** A Pareto chart of effects, main effect and interaction plots, and a table with the coefficients and p-values for each term in the model.

## 4.13 Weibull

**Purpose:** Analyze lifetime data to understand the reliability of a product. This method is ideal for datasets with two or more failures.

**How to Use:**

- Select the Failure Column and optionally a Censoring Column.
- Choose the Estimation Method (LSXY or MLE).
- Click on “Analyze”.

**Output:** A Weibull probability plot, parameters (Beta, Eta), percentiles, and a goodness-of-fit statistic.

## 4.14 Weibayes

**Purpose:** Estimate reliability with small sample sizes that have few or no failures. This method requires assuming a shape parameter (Beta) from prior knowledge.

**How to Use:**

- Enter the assumed Beta value (shape parameter).
- Add data points for failed and non-failed samples.

**Output:** A probability plot along with reliability estimates and confidence bounds based on the assumed Beta.

## 4.15 Warranty Prediction

**Purpose:** Simulate the risk of failure over time by comparing a load distribution with a strength distribution.

**How to Use:**

- Enter the parameters for the **Weibull (Strength)** and **Lognormal (Load)** distributions.
- Set a **Reference Year** for the analysis.
- Click on “Run simulation”

**Output:** An estimate of the failure probability (in % and PPM) for the reference year, a trend chart showing the risk over time, and a plot visualizing the overlap of the two distributions.

# 5 Exporting Results

Charts can be downloaded directly from the application for use in reports and presentations.

- Hover over a chart to reveal the Plotly mode bar at the top right.
- Click the **camera icon** to download the chart as a **PNG** or **SVG** file.

## 6 Version History (Changelog)

### **v1.2.3**

- Added Design of Experiments (DOE) module for 2-level full factorial experiments.
- Corrected German translations for DOE module terminology.
- Adjusted UI to hide the main input field for modules that do not use it (e.g., DOE).
- Fixed a bug where period/comma logic did not change on data input.
- Fixed a bug in the capability module that mirrored the x-axis in some cases.

### **v1.2.2**

- Histogram charts with faint lines.
- Correction of line values shown in charts.
- Change to the local file system to not save data from data sources.
- "Development Environment" item added.
- Axis scaling settings.

### **v1.2.1**

- Code adjustment for potential memory leaks.
- Fixed an issue where the 1-sample t-test results did not match between "Statistics Center" and "Minitab".

### **v1.2**

- Release version.